MILITARY SPECIFICATION

CANNULA, BRONCHIAL, PLASTIC, DISPOSABLE, JACKSON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sterile, disposable, bronchial suction catheters of Jackson design for use in bronchoscopy.

1.2 Classification. Cannulae shall be furnished in the following sizes, as indicated below (see 6.2).

Size

I - 9 Fr.
II - 12 Fr.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Set-up.

Military

MIL-S-36586	Sterilization Strip Test, Bacterial Spore.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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DRAWING

Defense Personnel Support Center (DPSC) Drawing No. 23305 Cannula, Bronchial, Plastic, Disposable, Jackson.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
Bureau of Biologics. Food and Drug Administration
Code of Federal Regulation Part 273. Minimum
Requirements (as applicable).

(Application for copies should be addressed to the Food and Drug Administration, 5600 Fishers Lane, Rockville, Maryland 20852.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (USP)

(Application for copies shall be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042.)

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

D792	Specific Gravity and Density of Plastics by Displacement, Tests for.
D638	Speed B, Tensile Properties of Plastics, Test for.

(Application for copies should be addressed to American Society for Testing and Materials, 1903 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Material. The tube shall be made of propylene-vinyl chloride copolymer. The connector shall be made of polyvinyl chloride plastic. The raw materials present in the composition of the tube and connector shall be in accordance with FDA regulatory provisions for materials in contact with food. In addition the tube shall comply with the biological safety requirements in the USP for Class II plastics.

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3.2 Style and design. The bronchial cannula shall consist of a single tube which at the proximal tenth of the length shall be bent 40° and at the proximal end shall have a connector slipped onto and cemented to the tube. The bronchial cannula shall comply with Defense Personnel Support Center (DPSC) Drawing No. 23305, dated 12 March 1974.

3.3 Specific gravity. The specific gravity of the material used in the tube shall be 1.33 ± .13 when tested according to the method indicated in 4.4.1.

3.4 Tensile strength and elongation. The tensile strength and percentage elongation shall be 5,000 psi and 90% respectively when tested as indicated in 4.4.2.

3.5 Flexural resistance. The maximum amplitude of deflection shall be 3/8 inch and 7/16 inch for the 9 Fr and 12 Fr, respectively, when tested according to the method described in 4.4.3.

3.6 Sterility. The bronchial cannulae shall be sterile when tested with a biological indicator as specified in 4.4.4.

3.6.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or as designated qualified alternate.

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When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa., 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Quality Assurance Representative for submission to the procuring activity.

3.6.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.7 Workmanship. Workmanship shall be first class throughout. The bronchial cannulae shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the tubing, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105 and as indicated in Table I. The unit of product for examination purposes shall be one cannula.

Table I. Sampling for Examination

Type of defect	Inspection level	AQL (percent defective)*
Major	S-2	1.5
Minor	S-1	1.5

*If acceptance number is not exceeded, lot shall be accepted. If rejection number is equaled or exceeded, inspection at Level I shall be performed to determine final acceptance or rejection.

4.2.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105 and as shown in Table II. The unit of product for test purposes shall be one cannula.

Table II. Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Specific gravity	3.3	4.4.1	S-1	2.5
Tensile strength	3.4	4.4.2	S-1	2.5
Elongation	3.4	4.4.2	S-1	2.5
Flexural resistance	3.5	4.4.3	S-1	2.5
Sterility	3.6	4.4.4	See 4.4.4	*

*Acceptance number shall be zero for all sample sizes.

4.2.3 Test samples. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

4.3 Examination. The bronchial cannulae shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Examination, however, shall not be restricted to defects listed in Table III. Any bronchial cannula in the sample, containing one or more defects shall be rejected, and if the number of defective bronchial cannula in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

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Table III. Classification of Defects

Categories	Defects
Major	
101	Cannula not free of roughness at distal end.
102	Cannula not free of roughness at areas other than distal tip.
103	Cannula kinked or otherwise obstructed.
104	Outside diameter of cannula out of tolerance.
105	Presence of foreign matter on exterior of cannula.
Minor	
201	Dimensions other than outside diameter of cannula out of tolerances.
202	Presence of foreign matter on interior of cannula.

4.3.1 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Examination shall be conducted as specified in Table IV, but not restricted to defects listed in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

Table IV. Examination of Preparation for Delivery

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Packing, packaging and materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper or missing.

4.3.1.1 Examination of closure and reinforcement for containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure and reinforcement shall be in accordance with the appendix to that specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.4.1 Specific gravity. Specific gravity shall be determined in accordance with ASTM Method D792, Method A.

4.4.2 Tensile strength and elongation. Tensile strength and percentage elongation shall be determined in accordance with ASTM Method D638, Rigid Tubes, Speed B. As an alternate, instead of the cannula, test specimens molded and cut to the dimensions indicated in ASTM Method D-638 may be used.

4.4.3 Flexural resistance. The method briefly is to have the tube span two supports, load the tube at the mid point, and determine the deflection from the horizontal. Specifically, the span and maximum deflection permitted as follows:

Size	Deflection, max.	Span
9 Fr	3/8 inch	6-1/2 inch
12 Fr	7/16 inch	11 inches

The deflection amplitude is measured by placing a straight edge across the two supports or uprights and measuring at the midpoint of the tube the vertical distance between the straight edge and the tube. A convenient height for the supports is 5 inches. The weight of the load which is attached to the tube specimen with a fine thread is 47.3 grams for both sizes of cannula.

4.4.4 Sterility test. The bronchial cannulae shall be tested for sterility in accordance with the method specified in the USP for parenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in Table II, Sampling Criteria for Tests.

4.4.4.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

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Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C
Dry Heat	B. stearothermophilus, or	55° - 65°C
	B. subtilis var. niger (or globigii)	30° - 35°C
Gas	B. subtilis var. niger (or globigii)	30° - 35°C
Radiation	B. pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.4.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.4 Sterility test and 4.4.4.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.4.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.4.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.4.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing ten cannulae, as specified constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Individual package. Each cannula shall be packaged in a sealed peel open type bag or a rigid plastic tray having a sealed, peel open type paper cover. The individual container shall be designed with tabs to facilitate opening. In addition, the container shall be capable of maintaining sterility of contents unless damaged or opened.

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5.1.2.2 Unit package. Ten individually packaged cannulae of one size only, shall be packaged in a snug fitting box constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Commercial colors are acceptable on unit packages and the closure shall be secured to prevent spilling of contents under normal handling.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twelve unit packages, containing cannulae of one size only, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as in the appendix of PPP-B-636.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B. Twelve unit packages containing cannulae of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall conform to method II as specified in the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.4 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.3 Marking,

5.3.1 Individual package. Each individual package shall be commercially identified and shall include markings required by the Federal Food, Drug and Cosmetic Act.

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5.3.2 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Sterility lot (control) number shall be shown. The method of sterilization (G-Gas, R-Radiation, C-Cold, S-Steam, D-Dry Heat Sterilization) shall be shown following the level of the packaging and date of packaging. Example: "A-3/74-G". Final sterilization shall be on or after indicated packaging date.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Sterility lot (control) number and the method of sterilization as specified in 5.3.2 shall be shown.

6. NOTES

6.1 Intended use. The cannulae are intended for use in ventilating the windpipe and airways of the lungs.

6.2 Ordering data. Purchasers should exercise any desire options offered herein, and the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

National Stock Nos.	Item Identification
6515-00-135-2614	CANNULA, BRONCHIAL, Plastic, Disposable, Jackson, 9 Fr., 10s (Size I)
6515-00-135-2629	CANNULA, BRONCHIAL, Plastic, Disposable, Jackson, 12 Fr., 10s (Size II)

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

6.4 Supersession data. This specification includes requirements of Purchase Descrption No. 1, dated 12 March 1974 for CANNULA, BRONCHIAL, PLASTIC, Disposable, Jackson.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1901
cs

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-37132

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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